

IMAC version 2.0: A L^AT_EX PACKAGE FOR WRITING PAPERS IN INTERNATIONAL MODAL ANALYSIS CONFERENCE (IMAC) FORMAT

Joseph C. Slater

Abstract

Just like in any other L^AT_EX document, you begin the abstract, make the title, and define the author as shown in the example. Not that IMAC requires that the author names be in bold, so you have to do this yourself. In addition, I haven't yet hacked the style definition for the title. Maybe I will later, but that's a minor inconvenience.

For feedback and support, please see the [IMAC style home page](#).

Keyword Keyword one • Keyword two • Keyword three

Nomenclature

$[M]$	matrix
$[^{\setminus} M,]$	diagonal matrix
$\{\Phi\}$	vector
(M_{11})	single element of $[M]$
$\ x\ _p$	p-norm of $\{x\}$
$\begin{bmatrix} [M]_{11} & : & [M]_{12} \\ \cdot & \cdot & \cdot \\ [M]_{21} & : & [M]_{22} \end{bmatrix}$	partitioned matrix

1 Changes

As of this release, versions are identified by release date (YYYY-MM-DD) rather than by an incrementing version number. Older entries below retain their original version numbers since precise individual dates for each of them could not be reliably recovered from the repository history; the closest available commit dates are noted.

2018-03-26: Initial public release (version 1.0, later also labeled 1.0.1/1.0.2), including the LPPL license and hyperlinks to the main document.

2021-12-08: (version 1.0.3) Fix minor typos in document.

2026-10-07: Realigned the style with the official IMAC Word Template (*Word Template.dotx*, distributed via the IMAC/SEM (Society for Experimental Mechanics) author toolkit) so that L^AT_EX and Word submissions now produce matching formatting. Specific changes:

- Switched body and heading fonts from sans-serif (Helvetica/Computer Modern Sans) to Times New Roman-compatible fonts (via mathptmx).
- Changed from a two-column US Letter layout to a single-column A4 layout, with margins matching the Word template (top 0.825in, bottom 1in, left/right 0.715in).
- Section, subsection, and subsubsection headings are now bold, sentence case, instead of bold, all-caps.
- The Abstract, References, Table, and Program labels are now bold, sentence case, instead of bold, all-caps.
- In-text citations are now plain, non-superscript, bracketed numbers (e.g. [1, 2, 3]), matching the Word template, instead of bracketed superscripts. Citation ranges are no longer compressed.

$[M]$	<code>\matr{M}</code>	matrix
$[\textit{M}]$	<code>\dmat{M}</code>	diagonal matrix
$\{\Phi\}$	<code>\vect{\Phi}</code>	vector
(M_{11})	<code>\elem{M}_{11}</code>	single element of $[M]$
$\ x\ _p$	<code>\pnorm{x}</code>	p-norm of $\{x\}$
$\begin{bmatrix} [M]_{11} & : & [M]_{12} \\ \dots & : & \dots \\ [M]_{21} & : & [M]_{22} \end{bmatrix}$	<code>\begin{bmatrix} \matr{M}_{11}&:&\matr{M}_{12}\\ \hdotsfor{2}{1}&:&\hdotsfor{2}{1}\\ \matr{M}_{21}&:&\matr{M}_{22} \end{bmatrix}</code>	partitioned matrix

Table 1: Defined macros

- Added a `\keywords` command to typeset a Keywords line after the abstract, matching the Word template.
- Paragraph spacing (`\parskip`) reduced to 8pt to match the Word template’s default paragraph spacing.
- Removed unused eleven/ten/eight-point `\font` definitions, the obsolete `helvet` package instructions, and the duplicate/stray package loading; added `\ProvidesPackage` for proper version reporting (now also date-based).
- Fixed a hyperlink color inconsistency: citations now use `citecolor=blue`, matching the rest of the document’s links (previously citations defaulted to `hyperref`’s green).

2 Typing Your Document

First thing to note is that the IMAC style indents the first line of each paragraph, including the paragraph immediately following a section header. You can override the L^AT_EX default of not indenting by using the command each time after a `\section` command or by using the package `indentfirst` which can be downloaded from any CTAN location.

The second thing which you may need to do is obtain and install the packages: `cite`, `ifthen`, `amsmath`, and `mathptmx` (for Times-compatible text and math fonts, matching the official Word template). These are all standard CTAN packages and should already be part of any complete L^AT_EX distribution.

The PDF® file `imac.pdf` can be viewed for comparison to the results you obtain from L^AT_EXing this document.

A few macros have been defined below for conforming to the IMAC [1] notation convention (See Table 1). If you use them, you can simply redefine the macros according to the journal requirements when the time comes for submission. I’ve defined only those that I thought were either difficult or do not conform to normal textbook standards. You may find it useful to define your own macros at the end of the file `imac.sty`, but please make sure that you don’t delete them the next time you update!

In Table 1 you’ll also see some code for making a partitioned matrix. Sorry, but I don’t know how to turn this into an environment. If you figure it out, let me know and I’ll incorporate it. The colons form the horizontal delimiters and the `\hdotsfor` command forms the vertical delimiters. The first argument represents a spacing of the dots, and the second required argument is the number of columns that the dotted line should span.

Please contact me at joseph.c.slater@gmail.com if you find bugs in this. I’ll do my best to fix them in a timely fashion. Please don’t contact me with respect to general L^AT_EX questions. For help with L^AT_EX please consult Lamport [2], Goossens et al. [3], and/or Kopka and Daly [4]. Additional resources are available through the T_EX newsgroup `comp.text.tex`, the CTAN archives at <http://www.ucc.ie/cgi-bin/ctan>, and the T_EXusers group (TUG) home page (<http://www.tug.org/>).

From here on is just some examples to give you some continuing examples of how to do things. The table would have been better off if I had used the `hhline` package.

3 Here is a section

In bladed disk assemblies, the disk acts as a coupling device between the blades. As the stiffness of the disk increases, blade coupling decreases. It has been shown that weak interblade coupling leads to high levels of mode localization when blades are mistuned.

There once was a figure here

Figure 1: Here is a figure.

Blade Number	Random 1	Random 2	Random 3
	10^{-3} Kg	10^{-3} Kg	10^{-3} Kg
1	0.0168	0.3343	0.3413
2	0.0260	0.2867	0.4431
3	0.2579	0.4529	0.3710

Table 2: Simple table.

3.1 This is a subsection

All models used in this study were variations of the symmetric, constant stiffness system referred to as the baseline model (Figure 1). Yada yada.

... and that's it! If you can't get it to work, I can be reached at `joseph.c.slater@gmail.com`.

Acknowledgments

Note that for the Acknowledgments you need to use the `\section` command in the starred form to avoid getting the section number.

Thanks to Leslie Lamport [2], Goossens, Mittelbach, and Samarin [3] and all others who've built $\LaTeX$ into what it is today.

Copyright

Copyright (C) 1998 Joseph C. Slater

References

- [1] **Ewins, D. J.**, *Notation for Modal Testing & Analysis*, Imperial College, London SW7 2BX, UK, 1993.
- [2] **Lamport, L.**, $\LaTeX$: A Document Preparation System, Addison-Wesley, 2nd edn., 1994.
- [3] **Goossens, M., Mittelbach, F. and Samarin, A.**, The $\LaTeX$ Companion, Addison-Wesley, 1994.
- [4] **Kopka, H. and Daly, P. W.**, A Guide to $\LaTeX$ 2e, Addison-Wesley, 1995.